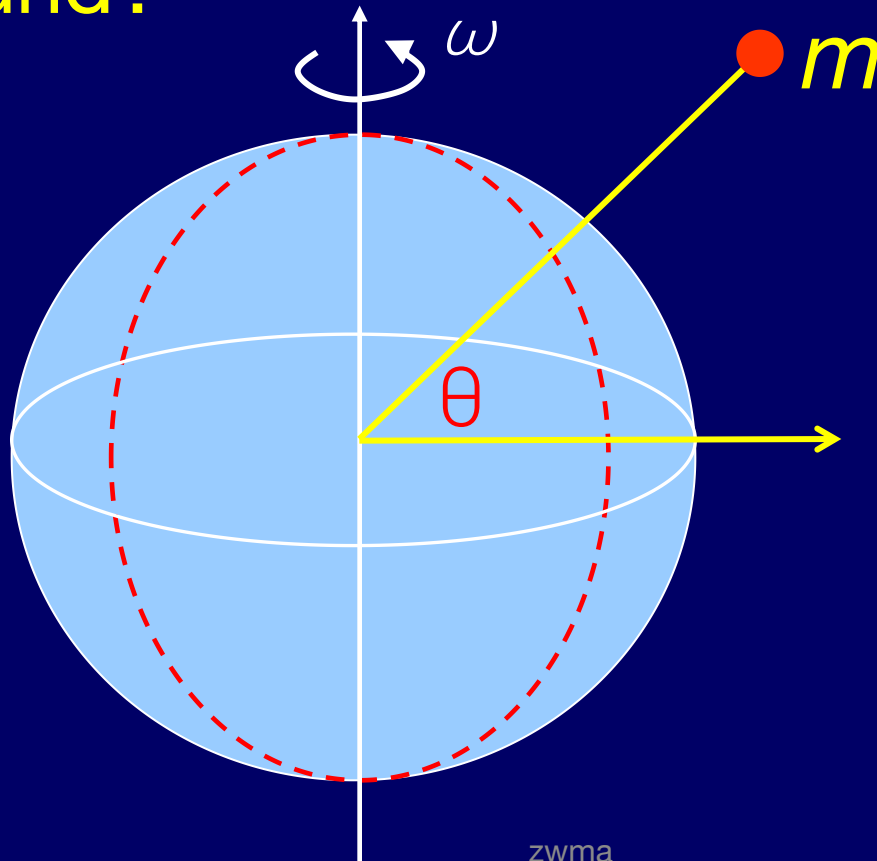


Quiz 3: A ball with mass m is dropped from height $h \ll R$, at the north latitude θ . How far to the east is the ball deflected, by the time it hits the ground?



Answer:

The angle between v and ω is $\theta + \pi/2$

The Coriolis force is $f = 2mv\omega \cos \theta$

Since $v = gt$, we have $a = 2gt\omega \cos \theta$

Thus, $v_{\text{east}} = gt^2\omega \cos \theta$

Or $d_{\text{east}} = gt^3\omega \cos \theta / 3$

Inserting $t = \sqrt{2h / g}$

$$d_{\text{east}} = \frac{2}{3} \sqrt{\frac{2h}{g}} \omega h \cos \theta$$